# Exit Ticket · reflect and retrieve

## Exit Ticket

Pipelining always reduces the number of stages required for an individual instruction.

## Use key vocabulary from memory

1. Explain clock speed in your own words.

2. Explain cache in your own words.

3. Explain pipeline in your own words.

## Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.





# Do Now · retrieval

## Clock speed, cache and pipelining

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

### Q1

What does the data bus transport during an instruction fetch?

### Q2

Which register holds the address of the next instruction?

### Q3

What is a dependency between tasks?

### Q4

A rising-edge D flip-flop initially has Q=0. D changes 0→1 between edges, then 1→0 just before the next rising edge. What is Q immediately after that edge? Justify.

# New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

## Core idea 1

Different performance factors — Clock speed indicates cycles per second; multiple cores can execute independent work; cache reduces some slower-memory access.

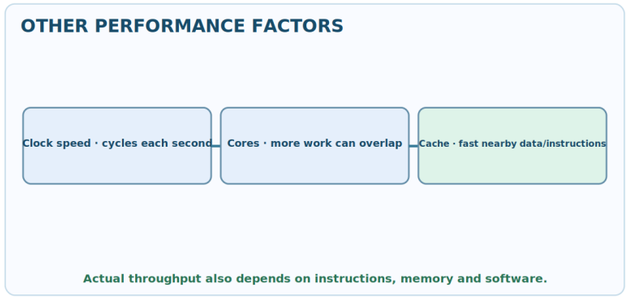
## Core idea 2

Overlap pipeline stages — Without overlap, three three-stage instructions need nine stage slots; an ideal pipeline finishes them by the fifth cycle.

## Core idea 3

Account for hazards — Branches, data dependencies and waiting for memory may cause stalls or pipeline flushing; pipelining does not make every instruction individually three times faster.

**Key visual · complete concept**




## Annotate the key visual

Explain which labels or features show the main process.





# You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 3

Explain why doubling clock speed does not necessarily double an application's performance. Include one pipeline hazard.

Check that you have used the technical terms and explained your decisions.

# You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

## Question 1

State two processor performance factors and explain one way each might improve throughput.

## Question 2

For a three-stage F–D–E pipeline with three independent instructions, write the stages occupied in cycles 1–5 and the completion cycle for I3.

Check that you have used the technical terms and explained your decisions.





# New Learning · applying the ideas

## Vocabulary · use the correct technical terms

clock speed — Number of processor clock cycles each second.

cache — Small, fast memory that reduces some main-memory accesses.

pipeline — Overlap stages of several instructions.

hazard — Dependency or uncertainty disrupting smooth pipelining.

## Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

# I Do · worked example

## Problem

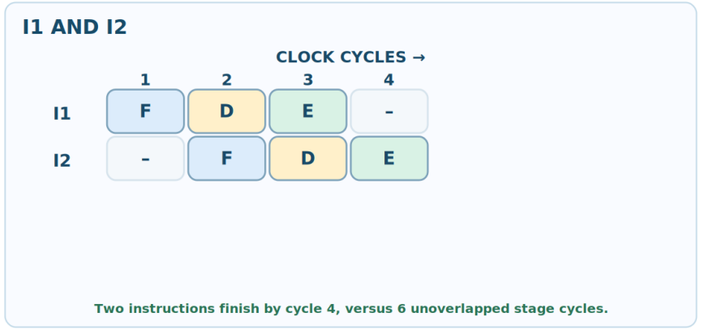
Trace two independent instructions through a 3-stage F–D–E pipeline.

## Follow the worked method

1. Predict first — Cycle 1 fetch I1. In cycle 2 decode I1 while fetching I2.

2. Explain and check — Cycle 3 executes I1 and decodes I2; I2 executes in cycle 4.

**Source model · final illustrated step**




Two instructions finish by cycle 4, versus 6 unoverlapped stage cycles.

## Why this method works

Identify the decision, rule or invariant that makes the method correct.





# We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

## Guided problem · TRY

Predict first

Begin a new fetch on each clock cycle, while older instructions advance to Decode or Execute.

## DISCUSS

Compare how you reached the answer. Where could someone make an error?

## IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.